

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 6

3 Computation Theory (ad260)

- (a) Define the collection of *primitive recursive functions*. [2 marks]

Primitive
recursive
functions

Answer: The class of primitive recursive functions is the smallest class of functions that includes the basic functions (zero^n , succ and proj_m^n) and is closed under composition and primitive recursion.

- (b) What does it mean to say that a function $f : \mathbb{N} \rightarrow \mathbb{N}$ is λ -definable. [2 marks]

lambda-calculus

Answer: f is λ -definable if there is a closed λ -term F that represents it. This means for all $x \in \mathbb{N}$,

$$F\underline{x} =_{\beta} \underline{f(x)},$$

where $\underline{n}$ denotes the Church numeral corresponding to $n \in \mathbb{N}$.

- (c) For each of these functions, show that it is primitive recursive and explain why it is λ -definable.

primitive and
partial recursive
functions,
lambda-calculus

- (i) $\text{fact}(x) \triangleq x!$; [7 marks]

Answer: Since $\text{fact}(0) = 1$ and $\text{fact}(x+1) = (x+1) \cdot \text{fact}(x)$, we have $\text{fact} = \rho^0(\text{succ} \circ \text{zero}^0, \text{mult} \circ [\text{succ} \circ \text{proj}_1^2, \text{proj}_2^2])$, where mult is as defined in the lectures.

For full marks, the answer should explain how mult is defined and also include the explanation in part (ii) below about λ -definability.

- (ii) $\text{cond}(x, y, z) \triangleq \begin{cases} y & \text{if } x = 0 \\ z & \text{otherwise} \end{cases}$ [7 marks]

Answer: $\text{cond}(x, y, z) = h(y, z, x)$ where h is defined as the function

$$\begin{aligned} h(y, z, 0) &= y \\ h(y, z, x+1) &= z \end{aligned}$$

So, cond can be defined as the function $h \circ [\text{proj}_2^3, \text{proj}_3^3, \text{proj}_1^3]$ where h is defined as $\rho^2(\text{proj}_1^2, \text{proj}_2^4)$.

It is shown in the lectures that every primitive recursive function is λ -definable. This is proved by showing that the basic functions are λ -definable and that the class of λ -definable functions is closed under composition and primitive recursion. For the latter, we use a fixpoint combinator.

- (d) Give an example of a function that is λ -definable but not primitive recursive. You do not need to give a proof of the fact. [2 marks]

primitive
recursive
functions,
computability

Answer: An example of such a function is the *Ackermann* function defined by $\text{ack}(0, y) = y + 1$, $\text{ack}(x + 1, 0) = \text{ack}(x, 1)$ and $\text{ack}(x + 1, y + 1) = \text{ack}(x, \text{ack}(x + 1, y))$.
